



NAVAL
POSTGRADUATE
SCHOOL

AE2440: Week 1

Modeling and Simulation

MATLAB as a calculator



Plan of the Day

- Review initial survey
- Review material from PMM Chapter 1
- Overview of the MATLAB and the default interface
- Scripts and Live Scripts
- Assignment 1 overview



Modeling and Simulation

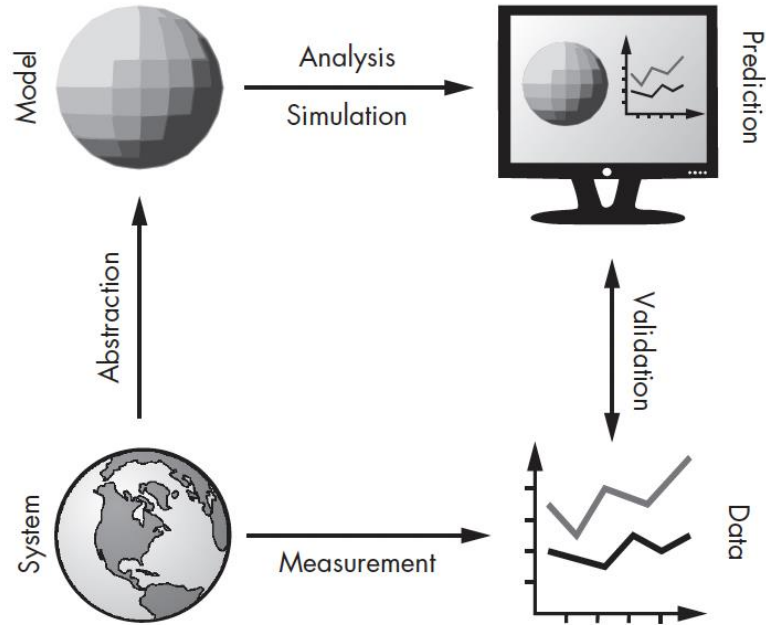
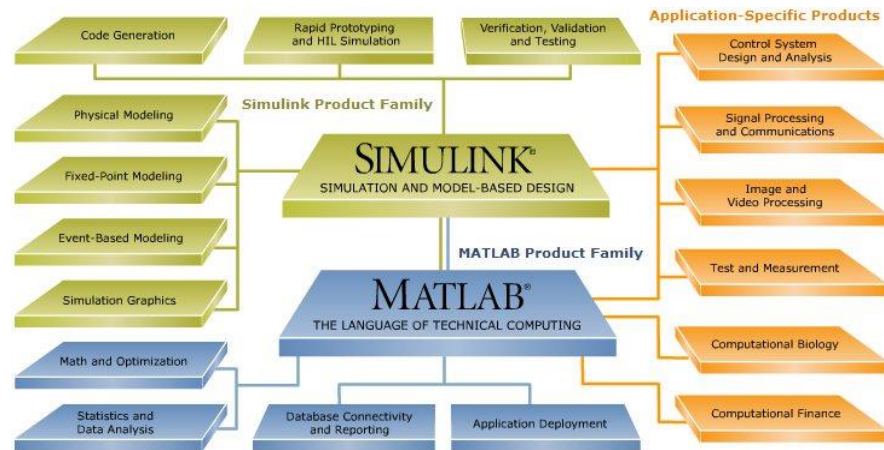


Figure 1.1: The modeling process



Why MATLAB/Simulink?

- **MATLAB:** High-level, interpreted programming environment for prototyping
- **Simulink:** A block diagram-based environment simulating dynamic systems.
- **Capabilities:** Constellation of "toolboxes". See [Products](#)
- **Documentation and Community:** Professional support, comprehensive documentation, large user-base.





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Workspace

Name	Git
bike_update.m	●
fibonacci_live.mlx	●
fibonacci_live.pdf	●
fibonacci1.m	●
mylivescript.mlx	●
mylivescript.pdf	●
mymyscript.m	●
swap.m	●

fibonacci1.m (Script)

Compute a numerical approximation of the nth Fibonacci number.

fibonacci1.m

```
1 % Compute a numerical approximation of the nth Fibonacci number.
2 % Precondition: you must assign a value to `n` before running
3 % this script.
4 % Postcondition: the result is stored in `ans`.
5
6 s5 = sqrt(5);
7 t1 = (1 + s5) / 2;
8 t2 = (1 - s5) / 2;
9 diff = t1^n - t2^n;
10 ans = diff / s5
11
```

Command Window

New to MATLAB? See resources for [Getting Started.](#)

>> fibonacci1

ans =

5.0000

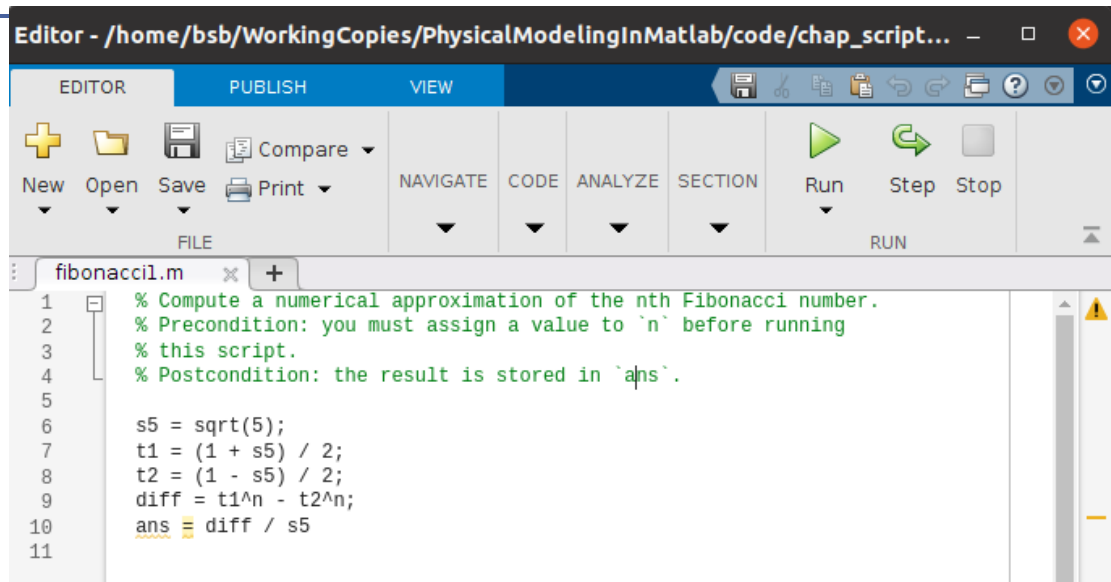
>>

Name	Value
ans	5.0000
diff	11.1803
n	5
s5	2.2361
t1	1.6180
t2	-0.6180



The way of the program

- Input
- Output
- Math
- Conditional Execution
- Repetition



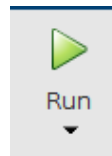
Programs and Scripts

- Are all scripts programs? Are all programs scripts?



MATLAB Script Execution

- *Run* entire program
 - From Command Window or from Editor



- Evaluate *selection*

- Run *Section*

```
5  
6  
7 s5 = sqrt(5);  
8 t1 = (1 + s5) / 2;  
9  
10 t2 = (1 - s5) / 2;  
11 diff = t1^n - t2^n;  
12 ans = diff / s5
```

Scope: the script variables are
in the main Workspace

```
fibonaccil.m * x +  
1 % Compute a numerical approximation of the nt  
2 % Precondition: you must assign a value to `n`  
3 % this script.  
4 % Postcondition: the result is stored in `ans`  
5  
6 %% Section 1  
7 s5 = sqrt(5);  
8 t1 = (1 + s5) / 2;  
9  
10 %% Section2  
11 t2 = (1 - s5) / 2;  
12 diff = t1^n - t2^n;  
13 ans = diff / s5
```



Getting the Code from the Book

- [Github](#)

bsb808 / **PhysicalModelingInMatlab**

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PhysicalModelingInMatlab Public

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This branch is **24 commits ahead of** [AllenDowney/PhysicalModelingInMatlab:master](#).

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Brian Bingham and Brian Bingham Adding notes de90ca3 · 2 days ago [348 Commits](#)

book	Release 1.0	last week
code	Release 1.0	last week
fonts	Updates from Overleaf	4 years ago
notes	Adding notes	2 days ago
.gitattributes	Adding live scripts and working on functions and fzero	last month



Live Scripts (AKA Notebooks)

- Live Script/Code = Code (programs and scripts + ...
Rich Text (formatted text, images, equations, etc.) + ...
Output (number, text, graphics, etc.)
- [Mathworks Gallery](#)
- Execution pattern is the same
 - Entire, selection or by *Section*
 - Variables accessible in main Workspace

The screenshot shows a MATLAB Live Script window with two tabs: 'fibonacci1.m' and 'fibonacci1_live.mlx'. The active tab is 'fibonacci1_live.mlx'. The script content is as follows:

The Fibonacci Sequence

The *Fibonacci sequence*, denoted F , is a sequence of numbers where each number is the sum of the previous two.

The following expression computes the Fibonacci number:

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$$

Assign a value for n

```
1  n = 10;
```

The result t is stored in 'ans'.

```
2  s5 = sqrt(5);
3  t1 = (1 + s5) / 2;
4  t2 = (1 - s5) / 2;
5  diff = t1^n - t2^n;
6  ans = diff / s5
```

ans = 55.0000



Assignment 1

- Content on Wiki

Assignment 1: Models and Scripts (25-1)

Created by Bingham, Brian (CIV), last modified on Sep 19, 2024

Completing assignments typically consists of writing programs (scripts, live scripts, function files, etc.) These files are submitted by uploading them to the AE2440 Sakai site under the appropriate assignment. I will then attempt to execute your programs. There are a few key things that will make this process run smoothly.

- To ensure that your run in a clean workspace (MATLAB environment), before submitting your final solution, execute the `clear` command and then run your program from an empty workspace. This will help avoid errors with undefined variables.
- Pay particular attention to the specified file names requested in your solution, including letter case. It is key that the file names be exactly as requested so that any automation on the evaluation end can run without a file-not-found error.
- When submitting live scripts, first run your Live Script to generate all specified numeric data, figures, etc. as inline output within your text/code. **Then** save the file. This will allow the instructor to review your solutions quickly, but also debug and re-run your code if necessary.

PMM stands for the textbook, Physical Modeling in MATLAB.

Problem 1: Penny

Write a script named `penny.m` to solve PMM Exercise 1.1.

It is good engineering practice to include comments with a description and units. I typically do this by including the units in square brackets, for example

```
% Time [s]
t = ...
```

- Submit via Sakai
 - Upload M-files
 - Naming matters
 - Case sensitive

<u>Assignment Title</u>	<u>Status</u>	<u>Open Date</u>	<u>Due Date</u>	<u>Last Modified</u>	<u>Modified Date</u>
<u>Assignment 1: Models and Scripts</u>	Not Started	Sep 19, 2024, 6:41 PM	Oct 4, 2024, 5:00 PM	Brian Bingham	Sep 27, 2024, 1:10 PM



Next Steps

- Wednesday: Read PMM Chapter 2: Scripts and Live Scripts
- Thursday: Lab, Working on Assignment 1